

Shunpeng Bai

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Education

Huazhong Agricultural University

Wuhan, China

B.S. IN BIOENGINEERING

September 2021 - June 2026

- **GPA:** 3.37/4.0
- **Main Courses:** Organic Chemistry, Calculus, Python Language Programming, Linear Algebra, Mathematical Modeling, College Physics, Probability Theory and Mathematical Statistics B, Biochemistry, Molecular Biology B, R Language and Scientific Plotting, Bioinformatics, Microbiology, Microbial Physiology, Microbial Pharmacology, Microbial Genetics and Breeding

Experience

Green Pesticide Chemistry Laboratory

HZAU, Wuhan, China

DIRECTED BY DR. HUAILONG TENG

Apr. 2022 - Oct. 2023

- “Synthesis of remote fluoroalkenyl ketones by photo-induced ring-opening addition of cyclic alkoxy radicals to fluorinated alkenes” D Du, H Peng, L He, S Bai, Z Li, H Teng - *Organic & Biomolecular Chemistry*, 2022
 - **Description:** Synthesized several compounds, and learned many experimental skills, including synthesis, isolation, characterization.

Biocatalysis Laboratory

HZAU, Wuhan, China

DIRECTED BY DR. SHUKE WU

Jun. 2021 - Oct. 2023

- Scientific research and training project was initiated titled “Chemico-enzyme cascade method for green and efficient preparation of Nepetalactone and its analogues”
 - **Description:** Directed evolution of P450-BM3 (CYP102A1) to obtain site selectivity for adipochain oxidation. I learned lots of technologies from DNA to protein, modifying protein, simulation *in silico* and detection of enzyme activity.

Natural Pharmaceutical Chemistry Laboratory

HZAU, Wuhan, China

DIRECTED BY ASSISTANT PROFESSOR SHENGZHEN XU

Oct. 2023 - Present

- “An organic small molecule fluorescent probe for nondestructive detection of Zn²⁺ in plants” J Zheng, S Li, S Bai, X Liu, R Sun, S Xu, W Qu-*Tetrahedron Letters*.2025 (Under Review)
 - **Description:** In this project, I undertook the work of theoretical calculation, including conformational search, DFT optimization and single point energy calculation for the probes before and after the combination. The reason why HOMO-LUMO gap becomes larger is explained. I learned a lot of theoretical calculation software and the use of computer cluster, and I wrote some scripts for convenient research.

Awards & Honours

- Served as a peer tutor for the chemistry course at Huazhong Agricultural University (2022.2-2022.7)
- Award for Excellence in the 14th “Huazhong Cup” College Students’ Mathematical Modeling competition
- Third Prize in the 15th “Huazhong Cup” College Students’ Mathematical Modeling competition
- Organic lecture titled “Understanding organic reaction from Markov addition reaction mechanism” for undergraduate student
- 2024 Gold Award in the Synthetic Biology Innovation Competition

Skills

Programming Python, R, Slurm, LaTeX, Bash, Git, Docker, Github Action

Spoken Languages Chinese (Native), English

Software Gaussian, Gromacs, Snapgene, AutoDock-vina, Multiwfn, Anaconda, VMD, MestReNova

Experimental skills Anhydrous and anaerobic reaction, Glove box, TCL plate, HPLC, Gas chromatography, Chromatographic column, Transformation, PCR, Molecular cloning, Protein express and purification, DNA sequencing analysis, Directed Evolution, Rational Design Enzyme, DNA AGE, Primer design, NMR spectrum analysis